

Statistics 250, Section 003, Quiz 2 (150 Points)

March 10, 1999, Dr. Jürgen Symanzik

Your Name: _____

Question 1: Short Answers (**40 Points**)

1. For $x_1 = 5, x_2 = -3, x_3 = 10, x_4 = -2, x_5 = 0, x_6 = 2$, and $n = 6$, determine the following sums: (**9 Points**)

$$\sum_{i=n/2}^n i \cdot x_i =$$

$$\sum_{i=3}^{n-2} x_{(i)} =$$

$$\sum_{i=n-2}^{n-1} \frac{x_{n-i}}{x_{(i)}} =$$

2. Calculate the z-scores (i.e., $z_i, i = 1, \dots, 6$) for the 6 values given in (1) above. It might be helpful to know that $\bar{x} = 2, \tilde{x} = 1$, and $s = 4.86$. **ROUND** to 2 significant digits (e.g., $2/3$ should be written as 0.67). (**6 Points**)

3. Determine the slope and the y-intercept of the lines whose equations are given as:
(8 Points)

(a) $5x - 5y = 20$

(b) $x + 8y - 16 = 0$

4. Indicate the slope and the y-intercept of the line that goes through the points $(x_1, y_1) = (5, 5)$ and $(x_2, y_2) = (2, 9)$.
(5 Points)

5. Let Z be a standard Normal variable, i.e., $Z \sim N(0, 1)$, and X be a Normal variable with mean $\mu = 1$ and variance $\sigma^2 = 4$, i.e., $X \sim N(1, 2^2)$. Determine the following:
(12 Points)

(a) $P(Z < 1.55) =$

(b) $P(-1.8 < Z < 2.0) =$

(c) $P(X < 1.55) =$

(d) $P(-1.8 < X < 2.0) =$

(e) Find a number $\#$ so that
 $P(Z < \#) = 0.10$

(f) Find a number $\#$ so that
 $P(X < \#) = 0.10$

Question 2: Scatterplot Matrix (**30 Points**)

The questions on the next page are based on the scatterplot matrix presented in Mark Monmonier's article "Geographic Brushing: Enhancing Exploratory Analysis of the Scatterplot Matrix". This scatterplot matrix has been reprinted below.

1. Label the (individual) scatterplot that shows the “Cable Penetration” on the horizontal (x-)axis and the “Per Capita Income” on the vertical (y-)axis with the letter “A”. **(3 Points)**

2. What is the range R of the “Cable Penetration” (in %)? **(3 Points)**

3. Which of these statements is correct/incorrect?
 - (a) The state with the highest “Per Capita Income” has a “Metropolitan Population” of less than 18%. **(4 Points)**

 - (b) The state with the highest “Cable Penetration” has the lowest “Metropolitan Population”. **(4 Points)**

 - (c) It is possible that Alaska is the state that has the highest “Cable Penetration”. **(4 Points)**

 - (d) The 6 New England states have “Cable Penetrations” that range among the 36 highest rates of “Cable Penetration” in the U.S. **(4 Points)**

 - (e) Texas is the state with the highest “Per Capita Income”. **(4 Points)**

 - (f) A low “Cable Penetration” implies a low “Per Capita Income”. **(4 Points)**

Question 3: Comprehensive Question on GMU Basketball (40 Points)

Below are the points GMU's Men's Basketball team has scored in their 29 regular games this season so far. This excludes exhibition games but includes final scores of overtime games. We want to determine some interesting statistical facts that relate to the entire season. However, the men's season is not finished yet since there will be at least one (and at most 6 ;-) more games in the NCAA tournament.

```
> bbmen
[1] 59 76 74 91 59 60 106 83 70 88 78 83 65 78 48 58 77 86 73
[20] 82 79 82 60 87 89 81 73 58 63
> sort(bbmen)
[1] 48 58 58 59 59 60 60 63 65 70 73 73 74 76 77 78 78 79 81
[20] 82 82 83 83 86 87 88 89 91 106
```

1. Calculate the mean, median, variance, and standard deviation (indicate whether this is based on a sample or entire population) of the men's scores. It might help to know that:

$$\sum_{i=1}^n x_i = 2166, \sum_{i=1}^n x_i^2 = 166470$$

(12 Points)

2. Numerically assess whether the data may originate from a $N(74.7, 13^2)$ distribution, i.e., a Normal distribution with mean 74.7 and standard deviation 13. **(These numbers are different from your answer in (1.) above!) (6 Points)**

3. Graphically assess whether the data originates from a Normal distribution. **(6 Points)**

4. Based on (2.) and (3.) above, does the data originate from a Normal distribution? Explain. (6 Points)

5. The GMU Women's Basketball team has scored the following number of points in their 27 regular games this season (which excludes exhibition games). The woman's season ended last week.

```
> bbwomen
[1] 64 46 76 76 77 99 79 57 62 69 66 69 58 71 70 79 62 75 73 45 84 51 83 50 78
[26] 60 51
> sort(bbwomen)
[1] 45 46 50 51 51 57 58 60 62 62 64 66 69 69 70 71 73 75 76 76 77 78 79 79 83
[26] 84 99
```

The GMU women scored 67.78 points in average with a standard deviation of 12.73 points this season. Since the team loses only one senior due to graduation, you can expect a very similar performance next season.

A good friend wants to bet with you about the outcome of the first regular game next season. We don't know the opponent yet but it is probably safe to assume a Normal distribution for the point distribution with the mean and standard deviation based on this season's results. You can choose among the following three bets and your friend, not knowing anything about statistics, will hold against any of these:

- (a) The team will score less than 60.5 points in their first game.
- (b) The team will score more than 72.5 points in their first game.
- (c) The team will score between 54.5 and 65.5 points in their first game.

Numerically evaluate these three options and argue which of them maximizes your chances to win this bet. Just indicating (a), (b), or (c) **DOES NOT** answer this question. (10 Points)

Question 4: Linear Regression & Correlation (**40 Points**)

When we introduced micromaps in class, we looked at a statistical display that contains information on the percentage of urban census tracts in a region and hazardous air pollutants in the same region. My claim was that there is a very close relationship between these two factors (however, we couldn't express this in terms of a regression line and correlation at that time).

Below is a scatterplot of the same data. Our explanatory (x -)variable in this case is the percentage of urban census tracts and our response (y -)variable is the average concentration of hazardous air pollutants in each state. We are looking at data of 48 states plus Washington, D.C., i.e, $n = 49$.

1. Fit a least squares (linear regression) line to the data. It might help to know that:

$$\sum_{i=1}^n x_i = 1850, \sum_{i=1}^n x_i^2 = 86405, \sum_{i=1}^n y_i = 1230, \sum_{i=1}^n y_i^2 = 34605, \sum_{i=1}^n x_i y_i = 53414$$

(20 Points)

2. What is a possible interpretation of the slope and y-intercept you calculated in (1) above? Explain. **(6 Points)**

3. Calculate Pearson's correlation coefficient r between x and y . How can we interpret this value for our given data set? **(6 Points)**

4. Based on your calculations in (1), what is the predicted concentration of air pollutants for a region that has 0%, 40%, 60%, 100% of urban census tracts. Which of these 4 predictions are more reliable, which are less reliable? Why? **(8 Points)**